Homework Submission

Student Name
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Solving the Quadratic Equation

We are given the following quadratic equation to solve:

$$x^2 + 6x + 3 = 0 \tag{1}$$

To find the roots, we use the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, where $a = 1$, $b = 6$, and $c = 3$:

$$x = \frac{-6 \pm \sqrt{6^2 - 4(1)(3)}}{2(1)} \tag{2}$$

Simplify the terms under the radical:

$$x = \frac{-6 \pm \sqrt{36 - 12}}{2} \tag{3}$$

$$x = \frac{-6 \pm \sqrt{24}}{2} \tag{4}$$

Since $\sqrt{24} = \sqrt{4 \times 6} = 2\sqrt{6}$, we can substitute and simplify:

$$x = \frac{-6 \pm 2\sqrt{6}}{2} \tag{5}$$

$$x = -3 \pm \sqrt{6} \tag{6}$$

Therefore, the final solutions are $x = -3 + \sqrt{6}$ and $x = -3 - \sqrt{6}$.